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Experimental study on the characteristics of capillary surface waves on a liquid film on an ultrasonically vibrated substrate

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Abstract

In this paper, we report experimental results on the propagation speed, wavelength, and frequency of capillary surface waves induced by imposing ultrasonic vibration on the substrate of a few millimeter thick water film. It is common knowledge that imposing a low frequency vertical vibration ($\sim$1–10^2 Hz) on a liquid body or layer results in bulk oscillation of the liquid and creation of standing surface waves with a frequency comparable to the excitation frequency (harmonic and subharmonic Faraday waves). Our experiments showed that the response of a liquid layer excited by horizontal and vertical ultrasonic frequency (20–68 kHz) is quite different, hypothetically because of a huge difference and mismatch between the characteristic times of vibration (small) and momentum diffusion (large) at high excitation frequencies. We observed that an excitation frequency of the order of 10^4 Hz results in the formation of traveling waves with frequencies of the order of 10^2 Hz, i.e. much smaller than the excitation frequency. In addition, we measured the speed of the Marangoni flow on the surface of a liquid film. We found that imposing ultrasonic vibration did not affect the speed of the Marangoni flow, corroborating that the ultrasonic vibration-induced traveling waves do not cause a bulk translational motion, and thus the flow on the surface is potential, whereas adjacent to the substrate a boundary layer forms. The studied problem is not only an intriguing fluid dynamics problem, it also has an existing application in the fabrication of thin coatings and solid films under ultrasonic excitation for emerging technologies, such as printed electronics and photovoltaics.

Keywords: liquid films, ultrasonic vibration, capillary surface waves, forced vibration, Marangoni motion

(Some figures may appear in colour only in the online journal)

1. Background

Imposing horizontal or vertical vibrations to a liquid layer may destabilize the free surface and induce gravity, gravity-capillary, or capillary surface waves, depending on the liquid depth and wavelength of the disturbances (Currie 2003). These surface waves, depending on the geometry and conditions of the problem, may be traveling or standing waves/patterns. Faraday (1831) first observed and investigated the standing waves, where he imposed low frequency piston-like vertical vibration on a liquid body in a vessel. While in the case of a low amplitude vibration, the liquid surface remains stable and flat and mimics the up and down motion of the oscillating basin, the surface may become unstable and a wave may form on the surface, if the amplitude becomes larger than a critical threshold. Following the experiments of Faraday and others, more than a century later, Benjamin and Ursell (1954) attempted to analyze the problem, theoretically, by writing the potential flow equations. They assumed that the vertical vibration term has an effect similar to that of the gravitational acceleration term, thus like most other authors, they treated the vertical vibration as an oscillating body force. By applying the linearized kinematic and dynamic boundary conditions on the liquid surface for small-amplitude waves and zero vertical velocity on the bottom surface of the liquid vessel, they obtained an equation for the velocity potential. Then, they performed a stability analysis based on the Mathieu's equation, to determine the regions where the liquid surface becomes unstable. Their analysis predicts that under certain conditions, the standing waves and indeed the entire body of fluid oscillate with a frequency half of the excitation frequency, in agreement with the basic experimental observations, although the theory predicts other modes of vibrations, as well. Since then, several authors have extended the analysis to more realistic cases by considering the viscous and damping effects, nonlinear behavior, and non-Newtonian flows, e.g. (Ezerskii *et al* 1986, Miles and Henderson 1990, Christiansen *et al* 1992, Edwards and Fauve 1994, Beyer and Friedrich 1995, Generalis and Nagata 1995, Kumar 1996, Müller *et al* 1997, Kumar 1999, Müller and Zimmermann 1999, Kumar 2002, Kumar and Matar 2002, 2004, Kumar and Tuckerman 2006, Milner 2006), among others. Fewer works have studied the problem of excitation of liquid layers by horizontal vibration, e.g. Yih (1968) and Or (1997).

The analysis of liquid layers subjected to vibration has been also extended to (thin) liquid films, e.g. (Kumar and Matar 2002, 2004, Matar *et al* 2004, Edwards and Fauve 2006, Shklyaev *et al* 2008, 2009, Benilov and Chugunova 2010, Bestehorn 2013, Bestehorn *et al* 2013, Garih *et al* 2013, Shklyaev *et al* 2015, Rahimzadeh and Eslamian 2017a). Oron *et al* (1997) and Craster and Matar (2009) have reviewed the literature on hydrodynamics of liquid films. Response of liquid films and thin liquid films to vibrational excitation has similarities and differences with that of deep liquid layers. In deep liquid layers with long waves, the surface waves are mainly due to the gravity, whereas in shallow liquid layers with short waves surface tension (capillary) and van der Waals forces (for ultrathin films) may become important. In most or all of the aforementioned works on thin liquid films, a long-wave lubrication approximation within a two-dimensional thin film has been assumed to simplify the governing equations and obtain regimes of stability and the formation and appearance of standing waves and patterns on the surface, e.g. (Shklyaev *et al* 2008, 2009, Bestehorn *et al* 2013, Rahimzadeh and Eslamian 2017a). While some authors, e.g. (Shklyaev *et al* 2008) have theoretically extended their analysis to high frequencies, it appears that the assumptions of lubrication theory break down at high frequencies, due to the formation of acoustic streaming in the flow and the significance of the boundary layer. In addition, in stability analysis of a liquid film, authors usually assume a capillary surface wave created by disturbances with an arbitrary wavenumber that satisfies the longwave approximation. However, in order to accurately and realistically evaluate the stability criteria and determine the window of stability of the film, the wavenumber of capillary waves should be known, *a priori* (Rahimzadeh and Eslamian 2017a). Thus, it appears that the present theories developed for thin films are inadequate for high frequency vibrations, simply because the longwave approximation is violated.

While most of the classical studies on vibration of liquid films and layer, mentioned above, consider a horizontal liquid layer on a substrate that is excited by a simple uniform vertical or horizontal vibration, in recent years a new application has emerged in which surface acoustic waves (SAWs) are employed to manipulate droplets and thin films for microfluidic applications. SAWs are generated by piezoelectric transducers at MHz frequencies and propagate and collide with a droplet or thin film as they travel laterally on the transducer surface (Friend and Yeo 2011, Yeo and Friend 2014, Rambach *et al* 2016). While the studies on SAW-driven thin films have benefited from the classical works on this film excited by the substrate from below, some works have employed a full set of governing equations to model acoustic streaming (Kolb and Nyborg 1956, Elder 1959, Manasseh 2016, Eslamian 2017) in the flow field, e.g. (Qi *et al* 2008, Tan *et al* 2010). In such numerical simulations, the liquid surface deformation can be captured, if a fine mesh is adopted. In particular, it has been observed that capillary surface waves that form on the film or droplet surface have frequencies several orders of magnitude smaller than the excitation frequency of the SAWs. This is interesting, because at a low frequency vibration (Faraday instability), the fluid usually follows the oscillations of the substrate or the vessel (vibration modulates with gravity) and the frequency of standing surface waves are comparable to that of the excitation frequency.

The aforementioned literature review and discussion reveals the demand for understanding the behavior and evaluating the characteristics of the surface waves that form on a liquid film on a vibrated surface at high frequencies. The practical motivation of this work was the application of ultrasonic vibration of the substrate for the fabrication of defect-free solution-processed organic coatings and thin film devices, e.g. (Wang and Eslamian 2016, Ahmadian-Yazdi and Eslamian 2018, Xiong *et al* 2018). In such applications, ultrasonic vibration ($\sim$40 kHz) has been imposed on a thin film of a solution, prepared by spin coating or other casting methods, to improve the leveling of the liquid film and the homogeneity and uniformity of the final solid film or coating, after solvent evaporation (Eslamian 2017). In this study, we considered water films of varying thicknesses in the millimeter range. The water films were excited by horizontal or vertical ultrasonic vibration at a range of 20–68 kHz. Using imaging, we observed and measured the characteristics of traveling surface waves. Some standing waves and patterns were also visualized, but not possible to characterize. We observed that in fact in thin films excited by kHz ultrasonic vibration imposed on their substrates, the capillary surface waves have a frequency much smaller than the excitation frequency, in agreement with the MHz SAW-driven thin films (Qi *et al* 2008, Tan *et al* 2010). We will discuss the reasoning behind this behavior in the following sections.

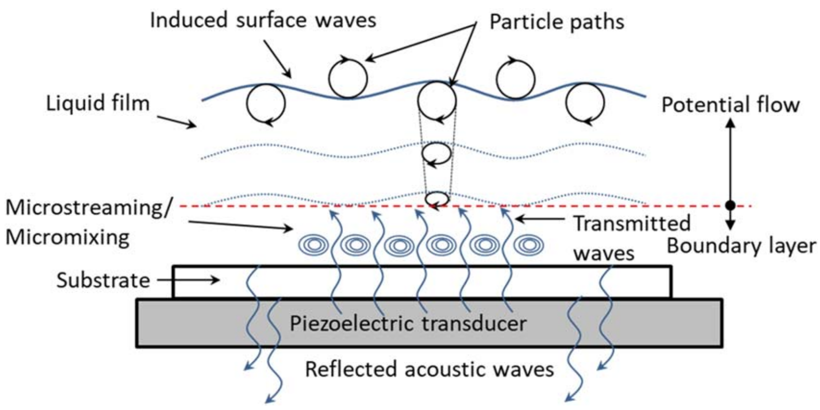

Figure 1. Effect of ultrasonic vibration excitation of a liquid film. The piezoelectric transducer may vibrate horizontally or vertically.

2. Experimental

Figure 1 illustrates a liquid film on a substrate, where the substrate is placed on an ultrasonically vibrating surface. The figure also shows some of the consequences of excitation of the film, particularly at large excitation frequencies, such as microstreaming and micromixing, appearance of traveling surface waves, formation of a boundary layer, and transmitted and reflected acoustic waves due to impedance mismatch between different media (Cheeke 2016). This figure represents the arrangement of our experiments. Our objective was to visualize and find the characteristics of the surface waves. For this purpose, we used distilled water to prepare liquid films with varying millimeter range thicknesses (2–9 mm) in a shallow (2 cm wall height) cylindrical vessel (Petri dish) with inner diameter of 13 cm. Few experiments with a water thickness below 3 mm were performed on 10 cm × 10 cm squared glass substrates. At such large substrates, the curvature was small and the water film was flat. At millimeter range liquid films on hydrophilic glass substrates, possible side effects of substrate properties, such as roughness, which is in nanometer range for glass substrates, are negligibly small. To cast a water film, we weighed and transferred a pre-calculated mass of water to the squared substrate or Petri dish with known surface area and spread it to obtain a leveled film. The contact lines were pinned to the edges of the squared substrates, and in the case of the Petri dish, the sidewalls upheld the water in the dish. In order to excite the water film by ultrasonic vibration, the squared substrate or the Petri dish were placed on an ultrasonically vibrating steel box. Inside each vibrating box, a disk-type Langevin ultrasonic transducer with a fixed resonant frequency of 20, 40, or 68 kHz and transducer input power of up to 50 W was mounted to transfer vertical ultrasonic vibrations to the top surface of the box and then to the water film. Each transducer was actuated by a signal generator at its particular resonant frequency (Yuhuan Clangsonic Ultrasonic Co., Ltd, Zhejiang, China). Horizontal vibration was achieved by turning the steel box by 90°. We used a 3D scanning vibrometer (PSV-500-V, Polytec, USA) to find the vibrational response of the ultrasonic transducers sensed on the surface of the steel boxes. At a transducer input power of 50 W, the 20, 40, and 68 kHz transducers generated a maximum displacement of 664, 7.8, and 0.5 nm, respectively. The details of measurements are provided elsewhere (Gholampour *et al* 2018).

In another set of experiments, a small amount of graphene flakes (Hengqiu Tech. Inc., China) was used as the seed particles to visualize the Marangoni motion. The graphene flakes (thickness of <5 nm and lamellar diameter of 10–50 μm) were dispersed on the surface of the water film. Then a 10 μl droplet of an aqueous surfactant solution, with a surface tension of 29.29 ± 0.16 mN m^{-1}, measured by a tensiometer (Theta-Lite, Biolin Scientific, Sweden), was dripped at the center of the water surface to induce a surface tension gradient and thus an outward radial Marangoni flow in the Petri dish. We conducted the experiments under various experimental conditions, i.e., no vibration and vibration at various frequencies.

We used a CMOS high speed camera (FASTCAM SA3 Model 120K, Photron, Japan) to capture the ultrasonically-induced capillary waves, and also to observe the Marangoni motion. The camera was set at 1500 frames per seconds and placed above the surface of the film at an angle of $\sim 80^\circ$ with respect to the film surface. The water film was illuminated with an LED light source. In order to find the wavelength and velocity of the traveling surface waves or seed particles in Marangoni motion, several successive images were analyzed by imageJ software, and the average values and the standard deviation of at least five measurements were obtained.

3. Results and discussion

On the surface of water films, we observed traveling waves toward the edges, as well as some moving wave patterns or standing waves. Figure 2 shows typical traveling and standing wave patterns, generated by a 40 kHz ultrasonic transducer on the surface of a water film in a Petri dish, where the effect of varying film thickness and transducer input power can be inferred, as well. The black lines are due to the application of adhesive tapes, used to attach the bottom of the dish to the surface of the vibrating steel box. Parts (a) and (b) of figure 2 show that at a constant film thickness, with decreasing the transducer input power, and therefore, the amplitude of vibration, both the standing and traveling waves attenuated. Likewise, increasing the film thickness, at a constant input power and amplitude resulted in attenuation of the standing waves, although the attenuated traveling waves are still visible (figures 2(c), (d)). Therefore, all parts of figure 2 collectively reveal that the standing and traveling waves appear on the surface, if the vibration power per film thickness is larger than a threshold. In other words, only a high intensity of imparted energy to the film can cause the formation of (visible) surface waves.

We also investigated the effect of varying the excitation frequency and the substrate geometry on the formation of surface waves. Figure 3 shows representative images of 3 mm water films formed on a Petri dish and squared substrates at excitation frequencies of 20, 40, and 68 kHz. This figure reveals that the patterns of the standing waves follow the geometry of the film platform and boundaries, in that the waves are circular in the case of Petri dish, whilst they have a checked pattern in the case of the squared substrate. Therefore, we attribute this behavior to different edge effects or boundary conditions, experienced by the liquid films. Figure 3 also shows that at constant input power of the transducer, increasing the frequency of the vibration (thus, decreasing the amplitude of vibration from 664 to 0.5 nm, when the frequency increases from 20 to 68 kHz) results in attenuation of the surface waves. We also applied higher frequency of 170 kHz, however, we were not able to detect a noticeable disturbance on the surface, supposedly due to a very small amplitude of vibration created by a 170 kHz transducer.

The traveling waves formed on a Petri dish are circular and easier to analyze, and the film thickness can be readily increased, as well. Thus, we performed systematic tests on water films with varying thickness formed in a Petri dish and measured characteristics of the traveling waves for both vertical and horizontal vibrations. The oscillation characteristics of

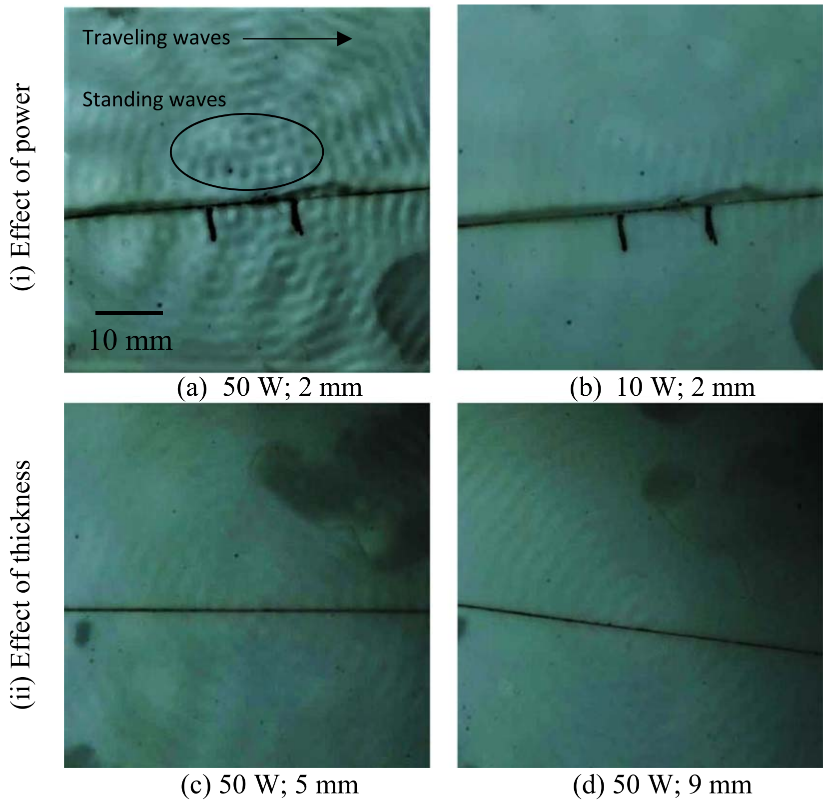

Figure 2. Standing and traveling waves created on water surface in a Petri dish excited by 40 kHz vertical vibration. (i) Effect of transducer input power: the film thickness is 2 mm, but the input power is 50 W in (a) and 10 W in (b). (ii) Effect of film thickness: the input power is 50 W, but the film thickness is 5 mm in (c) and 9 mm in (d).

the standing waves were not measurable from the images, because such oscillations were perpendicular to the plane of the images. The left and right columns of figure 4 show the wavelength, wave propagation velocity, and frequency (ratio of the velocity to the wavelength) of the traveling waves, due to horizontal and vertical ultrasonic vibration, respectively. We observe that for both vertical and horizontal vibrations, the wavelengths are ~2 mm, thus the waves are capillary waves (Currie 2003), i.e., gravity has no effect. We observed that the wavelengths only slightly vary with the film thickness and the excitation frequency. The graphs show that the velocity of the traveling waves is ~0.5 m s^{-1} and for the range of the ultrasonic frequencies studied here, the capillary wave velocity does not change with the excitation frequency, excitation direction (vertical or horizontal) or film thickness. The frequency of the induced surface waves is ~250 Hz (of the order of 10^2), while the frequency of excitation is of the order of 10^4 Hz, i.e. two orders of magnitude difference, consistent with the observations made by others for thin films interacted with SAWs (Qi *et al* 2008, Tan *et al* 2010). In the following, we will hypothetically explain the reason behind this phenomenon, by considering the characteristic times of vibration and momentum diffusion and the formation of a boundary layer and potential flow layer in the film.

Here we first show that water behaves like a potential flow on the surface, based on Marangoni motion experiments. We dripped a surfactant droplet on the surface of a 5 mm

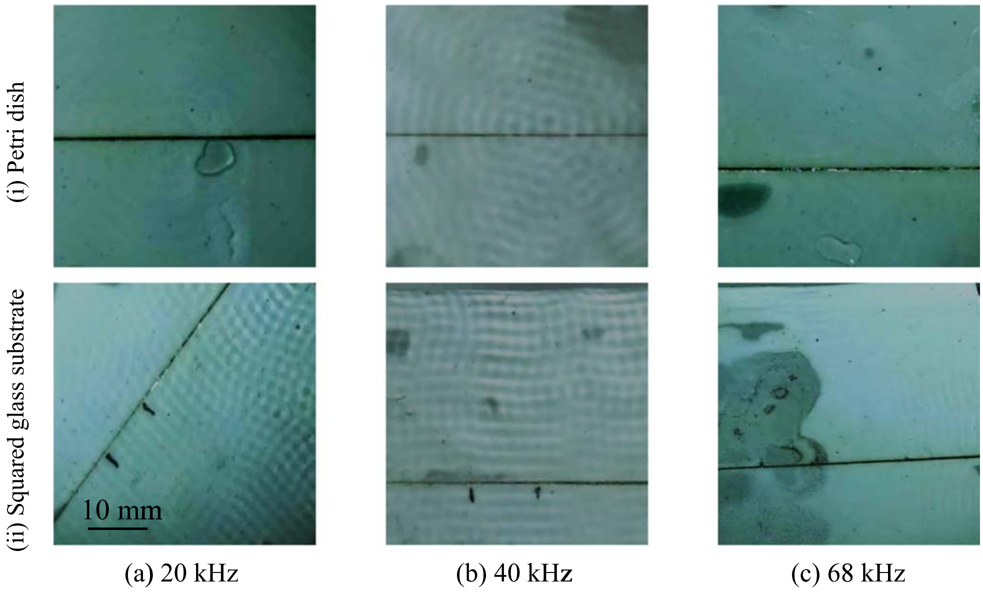

Figure 3. Surface waves appearing on a ~3 mm water film on (i) a Petri dish and (ii) a squared substrate, for vertical excitation frequency of (a) 20, (b) 40, and (c) 68 kHz at a vibration input power of 50 W. The main difference is that in (i) the waves travel radially outward, whereas in (ii) they travel parallel to the edges of the substrate.

thick water film seeded with graphene flakes formed in a Petri dish, to create an outward Marangoni flow. Upon release of the surfactant droplet, seed particles moved radially toward the boundaries, at the speed of the Marangoni flow. The spreading radius as a function of time is shown in figure 5, for the pristine film, and the films subjected to vertical vibration at 20, 40, and 68 kHz. The slope of the line is the velocity of the Marangoni flow (~0.8 m s^{-1}). The figure shows that imposing ultrasonic vibration, and thus creation of surface waves, with varying frequencies has no considerable effect on the propagation velocity of the seeds. Therefore, one can conclude that vibration-induced surface waves do not cause additional translational motion, i.e. the seed particles drift due to surface-tension driven Marangoni flow only, regardless of the imposed vibration. The potential flow hydrodynamic analysis of surface waves predicts that the particle paths on the liquid surface only locally circulate on elliptical paths for shallow liquid layers or on circular paths for deep liquid layers (Currie 2003) (figure 1). Thus, the Marangoni flow results of figure 5 ascertain that the flow on the surface is potential. On the other hand, near the substrate, the viscous effects are strong and a boundary layer forms. The viscous boundary layer thickness δ_v can be estimated by $\delta_v = (\mu/2\pi\rho f)^{1/2}$, where μ and ρ are the viscosity and density of the fluid, and f is the frequency of vibration (Morse and Ingard 1968, Tan *et al* 2010). δ_v for water at 20 °C at the frequencies used in this study is about 2 μm. Thus, our liquid films consist of a thin viscous or Stokes' boundary layer near the substrate and a thick potential outer flow above it. The ratio of the boundary layer to the film thicknesses is quite small and within the same order or magnitude for all excitation frequencies examined in this study. This may be responsible for a weak dependence of the characteristics of the surface waves to the film thickness and excitation frequency.

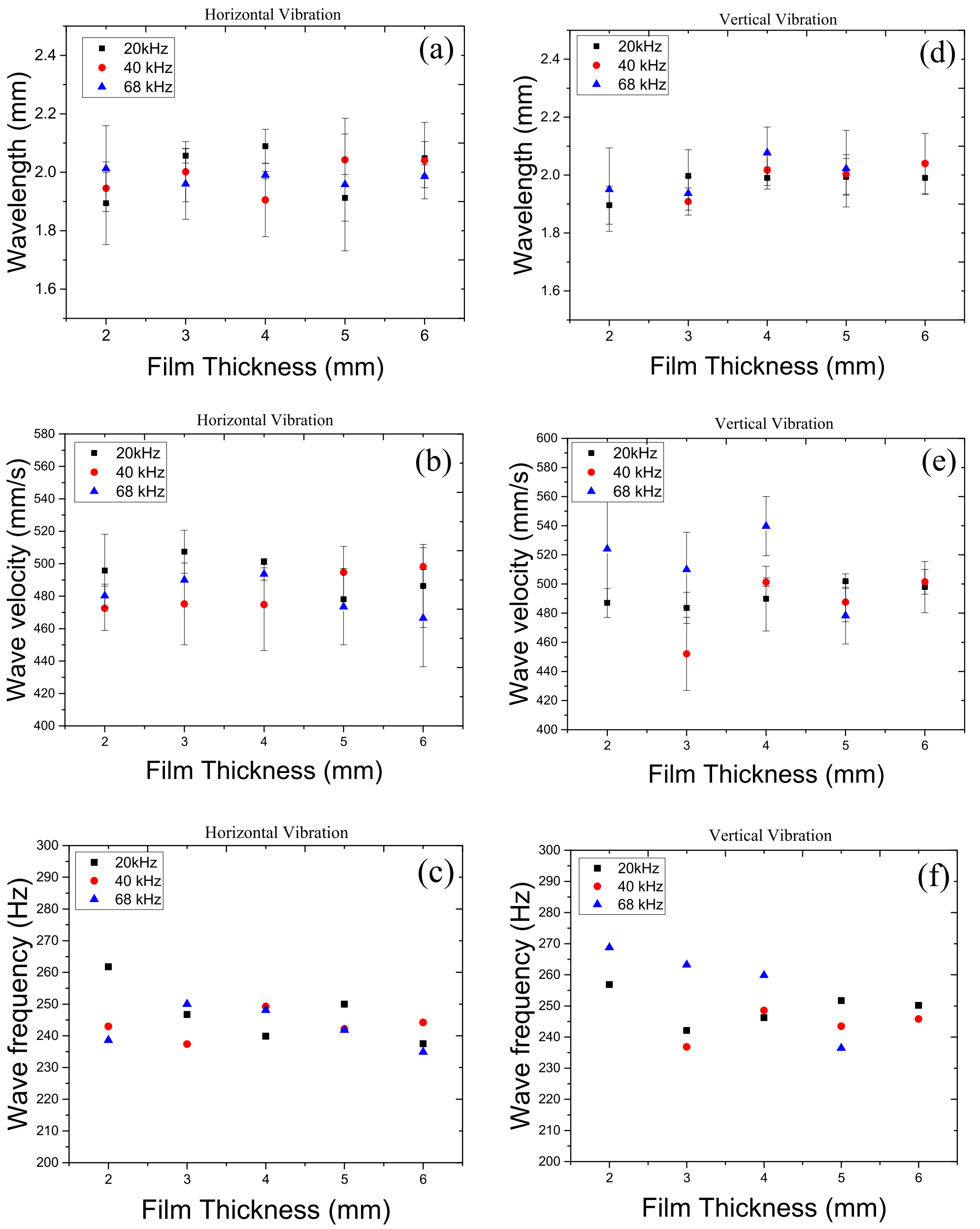

Figure 4. Measured wavelength (a), (d) and wave propagation velocity (b), (e), and their corresponding calculated wave frequency (c), (f) for traveling surface waves in a Petri dish, induced by vertical (left column) and horizontal (right column) ultrasonic vibrations at 50 W input power.

The existence of a thin viscous boundary layer, in which the acoustic energy is dissipated, and then a thicker potential outer layer atop the viscous layer may be responsible for a discontinuity in the flow behavior and a difference between the excitation frequency and the frequency of the induced surface waves. This may be interpreted in terms of the characteristic

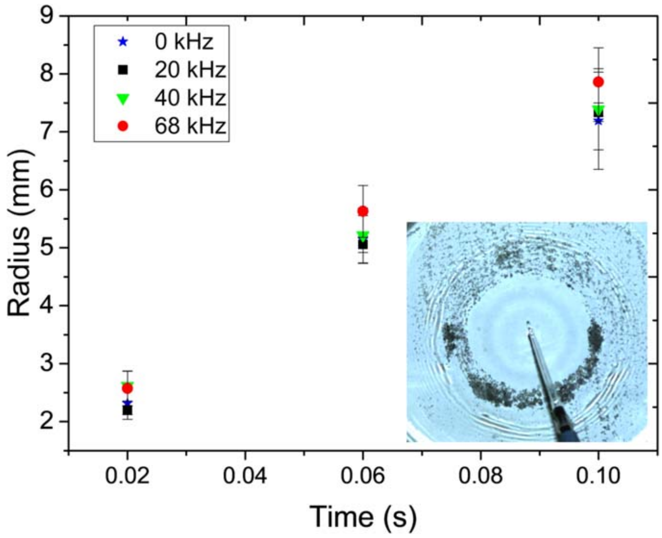

Figure 5. Time variation of propagation or expansion radius of seed particles due to the Marangoni flow under no vibration and several vertical ultrasonic vibration frequencies. The inset shows the picture of a film subjected to excitation at 68 kHz.

times of vibration $(1/f)$ and of viscous relaxation or momentum diffusion within the liquid film (h^2/υ), h being the film thickness and υ the liquid kinematic viscosity. For water and the range of film thicknesses and frequencies of this study, the viscous relaxation or momentum diffusion time is several orders of magnitude larger than the vibration characteristic time or period. For low frequency piston-like vibrations, these two characteristic times are of the same order of magnitude and the fluid can follow the vibrations. In contrast, for high frequency vibrations, the fluid cannot simply follow the vibrations, resulting in the formation of a boundary layer, viscous dissipation and streaming. Thus, the acoustic energy is dissipated in the boundary layer, which is indeed turbulent, and serves as a source of disturbance on the surface of the liquid film. Due to the chaotic nature of the turbulent boundary layer, a clear correlation between the excitation frequency and that of the surface waves is nonexistent. We should point out that we had observed similar behavior in the case of sessile droplets excited by ultrasonic vibrations. While at low frequency excitations of the substrate, the droplet typically oscillates with an frequency comparable in magnitude to the driving frequency (Lyubimov *et al* 2006), at ultrasonic vibrations (~40 kHz), the droplet oscillates with a much smaller frequency than the excitation frequency (Rahimzadeh and Eslamian 2017b).

As mentioned in the introduction, the present theoretical models that tackle the problem of thin film excitation by vertical, horizontal or tilted vibration, e.g. (Shklyaev *et al* 2008, 2009), are based on the lubrication approximation, in that a longwave with an arbitrary wavenumber is assumed on the surface. In other words, the wavenumber is an input to test the prediction power of the models in terms of the film stability, e.g. (Rahimzadeh and Eslamian 2017a). The experimental results of this work, although limited, provide a realistic estimate of the surface wave characteristics for a range of ultrasonic frequencies and film thicknesses. It is noted that for thinner films or low power vibrations, the surface waves are difficult to visualize with the current experimental capabilities. Performing numerical simulations, similar to the approach used in the SAW-driven thin films e.g. (Qi *et al* 2008, Tan *et al* 2010), can resolve the characteristics of the surface waves appearing on thin films excited by vertical or horizontal vibration. This means that a combination of acoustic equations and three-dimensional Navier–Stokes equations should be solved to capture the pressure waves, viscous effects, streaming, and transmission and reflection of acoustic waves at the interfaces, in order to determine the small-scale surface deformations and thus the characteristics of the surface waves, as well as all other flow parameters of interest.

4. Conclusions

In this work, we imposed ultrasonic vibration on the substrate of the thin films of water at frequencies of 20, 40, and 68 kHz with nanometer-ranged amplitudes and studied the formation of capillary surface waves. Analysis of the images revealed that, under the range of the parameters studied here, the frequency of the induced traveling surface waves was two orders of magnitude smaller than the excitation frequency. This finding is consistent with MHz SAW-driven experiments performed by others and substantiates the difference between the response of a liquid film to high frequency excitations and low frequency piston-like excitations. This is hypothetically because at low frequencies, the characteristic times of vibration and momentum diffusion from the substrate to the surface are comparable, whereas at high frequencies, the characteristic time of vibration is much smaller than that of the momentum diffusion. Therefore, the momentum due to acoustic energy is dissipated in the boundary layer and causes a disturbance on the surface with a frequency, which is not necessarily dependent on the excitation frequency.

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